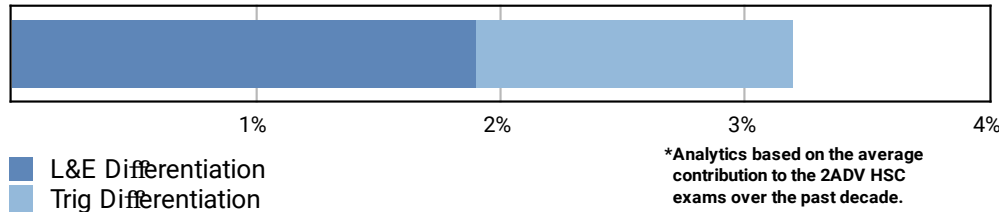


ADV: Calculus (Adv), C2 Differential Calculus (Adv)
Trig Differentiation (Y12)

Teacher: Denis Vlismas

Exam Equivalent Time: 52.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

C2 L&E and Trig Differentiation



HISTORICAL CONTRIBUTION

- C2 Log/Exponential and Trig Differentiation has contributed a healthy 3.2% to past Advanced exams, on average over the past decade.
- This topic has been split into two sub-topics for analysis purposes: 1-Log and Exponential Differentiation (1.9%), and 2-Trig Differentiation (1.3%).
- This analysis looks at Trig Differentiation.

HSC ANALYSIS - What to expect and common pitfalls

- This topic represents a low difficulty area where students should score highly. Again we flag that stand alone questions have been notably absent in the first two new syllabus exams.
- Trig Differentiation when examined prior to 2020 as a stand alone topic, has offered up very achievable band 3 difficulty marks.
- HSC Marker's Comments have noted that students who wrote out the quotient rule and all its inputs before substituting in values made less mistakes.
- Notwithstanding the omission in the 2020-21 papers, this topic is crucial gateway knowledge for numerous later Y12 topics and must be a core competency of every student.

Questions

1. Calculus, 2ADV C2 2016 HSC 5 MC

What is the derivative of $\ln(\cos x)$?

- (A) $-\sec x$
 (B) $-\tan x$
 (C) $\sec x$
 (D) $\tan x$

2. Calculus, 2ADV C4 2018 HSC 5 MC

What is the derivative of $\sin(\ln x)$, where $x > 0$?

- (A) $\cos\left(\frac{1}{x}\right)$
 (B) $\cos(\ln x)$
 (C) $\cos\left(\frac{\ln x}{x}\right)$
 (D) $\frac{\cos(\ln x)}{x}$

3. Calculus, 2ADV C2 2013 HSC 4 MC

What is the derivative of $\frac{x}{\cos x}$?

- (A) $\frac{\cos x + x \sin x}{\cos^2 x}$
 (B) $\frac{\cos x - x \sin x}{\cos^2 x}$
 (C) $\frac{x \sin x - \cos x}{\cos^2 x}$
 (D) $\frac{-x \sin x - \cos x}{\cos^2 x}$

4. Calculus, 2ADV C2 2019 HSC 11b

Differentiate $x^2 \sin x$. (2 marks)

5. Calculus, 2ADV C2 2012 HSC 12aii

Differentiate with respect to x .

$$\frac{\cos x}{x^2}. \quad (2 \text{ marks})$$

6. Calculus, 2ADV C2 2006 HSC 2aii

Differentiate $\frac{\sin x}{x+1}$ with respect to x . (2 marks)

7. Calculus, 2ADV C2 2004 HSC 3aii

Differentiate with respect to x :

$$(1 + \sin x)^5. \quad (2 \text{ marks})$$

8. Calculus, 2ADV C2 2006 HSC 2ai

Differentiate $x \tan x$ with respect to x . (2 marks)

9. Calculus, 2ADV C2 2007 HSC 2aii

Differentiate with respect to x :

$$(1 + \tan x)^{10}. \quad (2 \text{ marks})$$

10. Calculus, 2ADV C2 2009 HSC 2ai

Differentiate with respect to x :

$$x \sin x \quad (2 \text{ marks})$$

11. Calculus, 2ADV C2 2010 HSC 1e

Differentiate $x^2 \tan x$ with respect to x . (2 marks)

12. Calculus, 2ADV C2 2010 HSC 2a

Differentiate $\frac{\cos x}{x}$ with respect to x . (2 marks)

13. Calculus, 2ADV C2 2011 HSC 4a

Differentiate $\frac{x}{\sin x}$ with respect to x . (2 marks)

14. Calculus, 2ADV C2 2013 HSC 11c

Differentiate $(\sin x - 1)^8$. (2 marks)

15. Calculus, 2ADV C2 2017 HSC 11c

Differentiate $\frac{\sin x}{x}$. (2 marks)

16. Calculus, 2ADV C2 2008 HSC 2aiii

Differentiate with respect to x :

$$\frac{\sin x}{x+4}. \quad (2 \text{ marks})$$

17. Calculus, 2ADV C2 EQ-Bank 2

Differentiate with respect to x :

$$e^{\tan(2x)} \quad (2 \text{ marks})$$

18. Calculus, 2ADV C2 2019 MET1 1b

Let $f(x) = x^2 \cos(3x)$.

Find $f'\left(\frac{\pi}{3}\right)$. (2 marks)

19. Calculus, 2ADV C2 EQ-Bank 10

The function $f(\theta) = \sin^3(2\theta)$.

If $f'(\theta) = 6 \cos(2\theta) - 6 \cos^n(2\theta)$, find the value of n . (2 marks)

Worked Solutions

1. Calculus, 2ADV C2 2016 HSC 5 MC

$$y = \ln(\cos x)$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{-\sin x}{\cos x} \\ &= -\tan x \\ &\Rightarrow B\end{aligned}$$

2. Calculus, 2ADV C4 2018 HSC 5 MC

$$y = \sin(\ln x)$$

$$\begin{aligned}\frac{dy}{dx} &= \cos(\ln x) \times \frac{d}{dx}(\ln x) \\ &= \cos(\ln x) \times \frac{1}{x} \\ &= \frac{\cos(\ln x)}{x} \\ &\Rightarrow D\end{aligned}$$

3. Calculus, 2ADV C2 2013 HSC 4 MC

$$y = \frac{x}{\cos x}$$

$$\begin{aligned}\text{Let } u &= x & u' &= 1 \\ v &= \cos x & v' &= -\sin x\end{aligned}$$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \frac{vu' - uv'}{v^2} \\ &= \frac{\cos x \cdot 1 - x(-\sin x)}{(\cos x)^2} \\ &= \frac{\cos x + x \sin x}{\cos^2 x}\end{aligned}$$

$$\Rightarrow A$$

4. Calculus, 2ADV C2 2019 HSC 11b

Using the product rule:

$$\frac{d}{dx}(x^2 \sin x) = x^2 \cdot \cos x + 2x \sin x$$

5. Calculus, 2ADV C2 2012 HSC 12a(ii)

$$y = \frac{\cos x}{x^2}$$

$$\begin{aligned}u &= \cos x & u' &= -\sin x \\ v &= x^2 & v' &= 2x\end{aligned}$$

Using the quotient rule,

$$\begin{aligned}\frac{dy}{dx} &= \frac{u'v - v'u}{v^2} \\ &= \frac{-\sin x \cdot x^2 - 2x \cdot \cos x}{x^4} \\ &= \frac{-x \sin x - 2 \cos x}{x^3}\end{aligned}$$

6. Calculus, 2ADV C2 2006 HSC 2a(ii)

$$y = \frac{\sin x}{x + 1}$$

$$\text{Using } \frac{d}{dx}\left(\frac{u}{v}\right) = \frac{u'v - uv'}{v^2}$$

$$\begin{aligned}u &= \sin x & v &= x + 1 \\ u' &= \cos x & v' &= 1\end{aligned}$$

$$\therefore \frac{dy}{dx} = \frac{\cos x(x + 1) - \sin x}{(x + 1)^2}$$

7. Calculus, 2ADV C2 2004 HSC 3a(ii)

$$y = (1 + \sin x)^5$$

$$\begin{aligned}\frac{dy}{dx} &= 5(1 + \sin x)^4 \times \frac{d}{dx}(\sin x) \\ &= 5 \cos x (1 + \sin x)^4\end{aligned}$$

8. Calculus, 2ADV C2 2006 HSC 2ai

i. $y = x \tan x$

Using product rule

$$\frac{d}{dx}(uv) = u'v + uv'$$

$$\begin{aligned}\therefore \frac{dy}{dx} &= \tan x + x \times \sec^2 x \\ &= x \sec^2 x + \tan x\end{aligned}$$

9. Calculus, 2ADV C2 2007 HSC 2aii

$$y = (1 + \tan x)^{10}$$

$$\begin{aligned}\frac{dy}{dx} &= 10(1 + \tan x)^9 \times \frac{d}{dx}(\tan x) \\ &= 10 \sec^2 x (1 + \tan x)^9\end{aligned}$$

10. Calculus, 2ADV C2 2009 HSC 2ai

$$y = x \sin x$$

$$\begin{aligned}\frac{dy}{dx} &= x \cos x + \sin x \times 1 \\ &= x \cos x + \sin x\end{aligned}$$

11. Calculus, 2ADV C2 2010 HSC 1e

$$y = x^2 \tan x$$

Using product rule:

$$\frac{d}{dx}(uv) = u'v + uv'$$

$$\frac{dy}{dx} = 2x \tan x + x^2 \sec^2 x$$

COMMENT: There is no necessity to take out the common factor x in this case.

12. Calculus, 2ADV C2 2010 HSC 2a

$$y = \frac{\cos x}{x}$$

$$\begin{aligned}\text{Let } u &= \cos x & u' &= -\sin x \\ v &= x & v' &= 1\end{aligned}$$

Using quotient rule:

$$\begin{aligned}\frac{dy}{dx} &= \frac{u'v - uv'}{v^2} \\ &= \frac{-\sin x \cdot x - \cos x \cdot 1}{x^2} \\ &= \frac{-x \sin x - \cos x}{x^2}\end{aligned}$$

MARKER'S COMMENT: A significant number of students did not know the quotient rule and many found assistance by writing out u , u' , v , v' before going into calculations.

13. Calculus, 2ADV C2 2011 HSC 4a

$$y = \frac{x}{\sin x}$$

$$\begin{aligned}u &= x & u' &= 1 \\ v &= \sin x & v' &= \cos x\end{aligned}$$

$$\text{Using } \frac{d}{dx}(uv) = \frac{u'v - uv'}{v^2},$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{1 \cdot \sin x - x \cdot \cos x}{(\sin x)^2} \\ &= \frac{\sin x - x \cos x}{\sin^2 x}\end{aligned}$$

14. Calculus, 2ADV C2 2013 HSC 11c

$$y = (\sin x - 1)^8$$

$$\begin{aligned}\frac{dy}{dx} &= 8(\sin x - 1)^7 \times \frac{d}{dx}(\sin x - 1) \\ &= 8(\sin x - 1)^7 \times \cos x \\ &= 8 \cos x (\sin x - 1)^7\end{aligned}$$

15. Calculus, 2ADV C2 2017 HSC 11c

$$y = \frac{\sin x}{x}$$

$$\begin{aligned}\text{Let } u &= \sin x & u' &= \cos x \\ v &= x & v' &= 1\end{aligned}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{u'v - uv'}{v^2} \\ &= \frac{x \cos x - \sin x}{x^2}\end{aligned}$$

16. Calculus, 2ADV C2 2008 HSC 2aiii

$$y = \frac{\sin x}{x + 4}$$

$$\begin{aligned}u &= \sin x & u' &= \cos x \\ v &= x + 4 & v' &= 1\end{aligned}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{u'v - uv'}{v^2} \\ &= \frac{\cos x(x + 4) - \sin x}{(x + 4)^2}\end{aligned}$$

17. Calculus, 2ADV C2 EQ-Bank 2

$$y = e^{\tan(2x)}$$

$$\begin{aligned}\frac{dy}{dx} &= \frac{d}{dx} \tan(2x) \times e^{\tan(2x)} \\ &= 2 \sec^2(2x) \cdot e^{\tan(2x)}\end{aligned}$$

18. Calculus, 2ADV C2 2019 MET1 1b

$$f(x) = x^2 \cos 3x$$

$$f'(x) = x^2 \cdot 3(-\sin 3x) + 2x \cos 3x$$

$$\begin{aligned}f'\left(\frac{\pi}{3}\right) &= \left(\frac{\pi}{3}\right)^2 \cdot 3(-\sin \pi) + 2\left(\frac{\pi}{3}\right) \cos \pi \\ &= -\frac{2\pi}{3}\end{aligned}$$

19. Calculus, 2ADV C2 EQ-Bank 10

$$\begin{aligned}f(\theta) &= \sin^3(2\theta) \\ &= (\sin(2\theta))^3\end{aligned}$$

$$\begin{aligned}f'(\theta) &= 3 \times 2 \cos(2\theta) \times \sin^2(2\theta) \\ &= 6 \cos(2\theta)(1 - \cos^2(2\theta)) \\ &= 6 \cos(2\theta) - 6 \cos^3(2\theta)\end{aligned}$$

$$\therefore n = 3$$